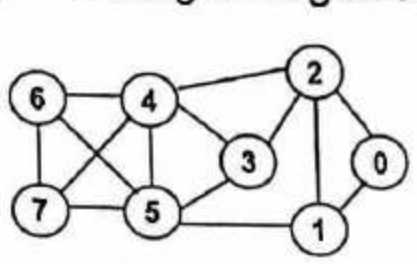
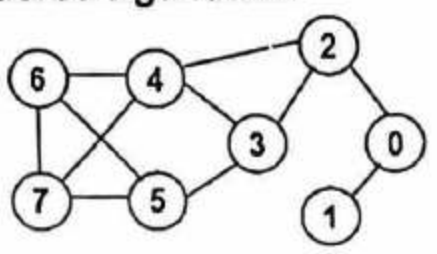
End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.TECH, Semester:8th
(May, 2025)

Subject Code: BAI 404	**Subject: Social Media Analytics**
Time :3 Hours	**Maximum Marks :60**

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8=20)**	**CO Mapping**
	a) Briefly differentiate how small organizations and large organizations approach SMA		**CO1**
	b) Give one basic measure used to quantify influence in a social network.		**CO1**
	c) Name two common applications of Natural Language Processing (NLP) for micro-text analysis on social media.		**CO2**
	d) What do you understand by influence maximization in a network?		**CO2**
	e) In random graph models, what does the parameter "p" usually represent?		**CO3**
	f) Give one example of how A/B testing might be used to improve a website's performance.		**CO3**
	g) What is influence maximization in the context of social media?		**CO4**
	h) Define reach and engagement in the context of Facebook Analytics.		**CO4**
	UNIT I		**CO Mapping**
Q2	Suppose you have a small social network with five users. Each user has certain connections. • Illustrate how you would represent this network using a graph and its corresponding adjacency matrix. • Define any two basic measures (e.g., degree centrality, clustering coefficient) you would use to identify an influential user and justify your choice.	**(10)**	**CO1, CO2**
Q3	Discuss the need for Social Media Analytics in organizations, highlighting the differences in usage for small versus large organizations. Illustrate how SMA can be applied in two distinct areas (e.g., marketing, customer service, crisis management).	**(10)**	**CO2**
	UNIT II		**CO Mapping**
Q4	You are given clickstream data of a website. Design a Python-based pseudocode or actual code to count the number of clicks per user session. Also explain how A/B testing can help optimize the user interface.	**(10)**	**CO3**
Q5	Every year, thousands of research papers are published worldwide. Suppose a researcher wants to work in a particular research area and explore some research papers in that domain. As millions of scientific papers are available in that domain, how will researchers select research papers? Use an appropriate network model and implement the above scenario. To implement the above behavior, consider that the maximum number of nodes in the network cannot exceed 10. Initially, a small network (m0) with 3 nodes is given, where each node is connected with the remaining nodes (the degree of each node is 2). Suppose a new node (researcher) wants to join the network with an incoming degree (m) of 2 (wants to read 2 papers). How will it choose the nodes to which it intends to connect? Please note that the final network will contain 10 nodes.	**(10)**	**CO3, CO4**

	UNIT III		CO Mapping
Q6	A social campaign on Facebook reached 15,000 people and had an engagement rate of 6.5%. (i) Calculate the number of people who are engaged. (ii) If the goal was a 10% engagement rate, how many more people needed to engage? (iii) Suggest improvements to meet the campaign goal.	(10)	CO3
Q7	a. Describe any three key metrics provided by Facebook Insights (e.g., demographics, reach, engagement). Why are these metrics important for page administrators? b. Propose a strategy to measure and analyze a social media campaign on Facebook, from defining goals to evaluating outcomes.	(10)	CO3
	UNIT IV		**CO Mapping**
Q8	Compare the efficacy of the Adamic-Adar Index, Jaccard Coefficient, Common Neighbors, and Preferential Attachment link prediction algorithms for the following network and determine the best performing among all considered algorithms.  Actual Network  Network with removal of 3 links	(10)	CO4
Q9	Write a Python program that crawls Twitter data from Twitter now known as X. Write all steps and also demonstrate how to visualize hashtags using word clouds or bar charts. Also, mention the steps for exploring and analyzing such data.	(10)	CO3, CO4